

第一届“认证杯”数学中国

数学建模国际赛 承 诺 书

我们仔细阅读了第一届“认证杯”数学中国数学建模国际赛的竞赛规则。

我们完全明白，在竞赛开始后参赛队员不能以任何方式（包括电话、电子邮件、网上咨询等）与队外的任何人（包括指导教师）研究、讨论与赛题有关的问题。

我们知道，抄袭别人的成果是违反竞赛规则的，如果引用别人的成果或其他公开的资料（包括网上查到的资料），必须按照规定的参考文献的表述方式在正文引用处和参考文献中明确列出。

我们郑重承诺，严格遵守竞赛规则，以保证竞赛的公正、公平性。如有违反竞赛规则的行为，我们将受到严肃处理。

我们允许数学中国网站(www.madio.net)公布论文，以供网友之间学习交流，数学中国网站以非商业目的的论文交流不需要提前取得我们的同意。

我们的参赛队号为：

我们选择的题目是：

参赛队员（签名）：

队员 1：夏裕俊

队员 2：李霖

队员 3：冯斌

参赛队教练员（签名）：

第一届“认证杯”数学中国

数学建模国际赛 编号专用页

参赛队伍的参赛队号：（请各个参赛队提前填写好）：1025

竞赛统一编号（由竞赛组委会送至评委团前编号）：

竞赛评阅编号（由竞赛评委团评阅前进行编号）：

Optimal Motion Designs for Robots with Coffee under Different Road Conditions

Abstract:

In this report, we build three progressive models: the quasi-static model, the spherical pendulum model, and the biped robot model. We find that the influencing factors of liquid sloshing are the acceleration of walking and the cup tilt angle, which can be used to characterize our motion designs for the walking process.

In model 1, by the quasi-static assumption, we find that the spill amount is minimal when the acceleration and the cup tilt angle satisfy the expression below. We then preliminarily devise a feasible variation of the acceleration and the cup tilt angle.

$$\theta = \tan^{-1}(a / g)$$

In model 2, we let the liquid surface be equivalent to the spherical pendulum. By introducing random turbulence factors, we simulate the walking with our initial motion design. And we calculate the spill amount of coffee, which is only 3.3% of the whole cup volume. Obviously the result is so idealized that it needs to be improved.

In model 3, we take into account the disconnection of force during walking and build a more accurate model. Modifying our motion design with the introduced walking frequency, we finally get the spill amount approximating to 10% of the cup volume after simulation for 1000 times. The ultimate result corresponds to the actual situation well.

Then we apply the models mentioned under three different road conditions (straight roads, roads with slopes and roads with corners). Aiming at the least spill amount of coffee, we get the optimal acceleration design (i.e. the motion design) for the three road conditions. The acceleration curve of the walking process is the radical type, the linear-radical mixed type and the quadric type, respectively.

For a cylindrical cup whose dimension (height H , radius R) is given, we find the following expression of the walk speed when the liquid system resonates. All the results derived from this expression are in accordance with our simulation.

$$\bar{v} = \frac{P}{2\pi} \sqrt{\frac{g\varepsilon_{mn}}{R} \tanh\left(\varepsilon_{mn} \frac{H}{R}\right)}$$

The major innovation of this report is: we simplify the problem by using the quasi-static model and the spherical pendulum model, which is also easy for simulation; we design many road conditions and acceleration types, making our analysis more considerate and convincing.

Key words: liquid sloshing quasi-static process spherical pendulum
biped robot acceleration type cup tilt angle

Description of problem

In our busy lives, sometimes we have to walk with a cup of coffee. It is quite common that coffee spills out of the cup during our walking process. When it comes to a robot walking with a cup of coffee, our goal is to keep coffee within the cup as much as possible by devising the motion form of the robot as well as the way of holding the cup.

Assumptions

- The road is flat with no holes or obstacles.
- Neglect the weather factors, e.g. wind and earthquakes.
- The length of one step is a constant.
- No slips or falls exist.
- People walk straight on straight road.
- No breakdown or time hysteresis during the walking of a robot.

Definitions

θ	Cup tilt angle, the angle between the cup and the horizontal plane
φ	Surface angle, the angle between the free surface of the liquid and the horizontal plane
l	Equivalent length of the pendulum
ω	Rotation frequency
R	Radius of the cup
H	Height of the cup
h	Height of the remaining coffee
η	Viscosity of coffee
ρ	Density of coffee
v	Average velocity
S	Distance
t	Total time of motion

T	Period of walking
Tc	Time of the speed variation on the slope
α	Dip angle of the slope
β	Time coefficient of proportionality at the uniform speed stage on the slope
ψ	Angle which describes the deviation of the pendulum
K	Turbulence factor

Model Analysis

Factors involving the spill amount

1.The size of the coffee cup.

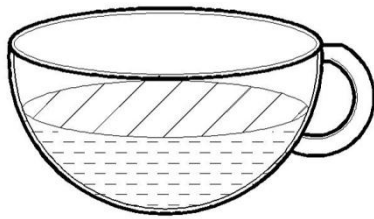


Figure 1. The cylindrical cup.

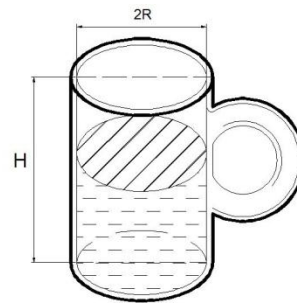


Figure 2. The truncated-cone cup.

As shown in Figure 1 and Figure 2, there are two common types of coffee cups, the cylindrical shape and the truncated-cone shape. Considering that what we care most is the liquid surface in the container, we study the cylindrical-shape cup only, for the two shapes of cup are almost the same when we just care about the liquid surface.

The sloshing of the coffee in the cup can be considered as the liquid oscillation, thus having its own natural frequency. The frequency, demonstrated by the following deduction, is related to the height and diameter of the cup.

Assuming that coffee is frictionless, vorticity-free, and incompressible liquid, the natural frequency of its oscillation in a cylindrical cup with a free liquid surface is well known from liquid sloshing engineering^{[1],[2],[6]}:

$$\omega_{mn}^2 = \frac{g \varepsilon_{mn}}{R} \tanh\left(\varepsilon_{mn} \frac{H}{R}\right) \left[1 + \frac{\sigma}{\rho g} \left(\frac{\varepsilon_{mn}}{R}\right)^2\right] \quad (1)$$

where $m = 0, 1, 2, \dots$ and $n = 1, 2, \dots$. In this equation H is the liquid height, ρ its

density, R the cup radius, σ the surface tension, and g the gravity acceleration. The values m_n are the roots of the first derivative $J'_m(\varepsilon) = 0$ of the m th-order Bessel function of the first kind; numerical values are $\varepsilon_{11} = 1.841$, $\varepsilon_{12} = 5.331$, $\varepsilon_{31} = 8.536$, $\dots$, $\varepsilon_{1,(n+1)} = \varepsilon_{1,n} + \pi$. For a typical common size of a coffee cup, $R \approx 3.5\text{cm}$ and $H \approx 10\text{cm}$, the surface tension σ effect is negligible and the angular natural frequency is $\omega_{11} = \omega_{nat} \sim 23 \text{ rad/s}$, so that the corresponding frequency is $\nu_{11} = \nu_{nat} \sim 3.65 \text{ Hz}$. In general, coffee cups vary in size roughly from $(R, H) = (2.5, 5.7)$ to $(6.7, 8.9)$ cm for espresso to cappuccino type mugs, which gives natural frequencies in the range 2.6~4.3 Hz.

There is also a frequency when people walking, thus the motion of a human acts on a driving force to the container. For one whose length of a step is p , we can find a certain speed $\bar{v}$ of which the corresponding walking frequency can generate resonance in the cup.

From the literature^[3], we have the following expression:

$$\bar{v} = \frac{p}{2\pi} \sqrt{\frac{g\varepsilon_{mn}}{R} \tanh\left(\varepsilon_{mn} \frac{H}{R}\right)}$$

2. The road condition

It is a common sense that the rougher the road condition is, the more violent sloshing the coffee has. Therefore, the spill amount of coffee in this situation is more than that of a flat road. In addition, if the road is with corners, it will generate more complex sloshing when making a turn, as the oscillation distributes on different directions under this condition.

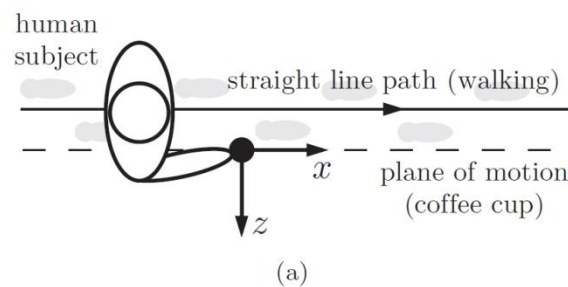


Figure 3. Effects of the road condition.

3. The stability of forces on the coffee cup

As the mass of the cup with coffee is almost invariable, we can use acceleration a and the changing rate of it, da/dt to represent the motion stability, since acceleration is a reflection of force. Supposing the cup moves with a uniform acceleration a , according to D'Alembert's principle, besides the gravity, there is an imaginary inertia force [see figure 4] acting on the coffee in the cup.

Thus we have:

$$X = a, Y = 0, Z = -g \quad (2)$$

Substituting (2) in the equilibrium differential equation of static fluid, $dp = \rho(Xdx + Ydy + Zdz)$ where X, Y, Z are the mass force on three orthogonal directions, respectively, we get:

$$dp = \rho(Xdx + Zdz) = \rho a dx - \rho g dz \quad (3)$$

Then integrating the formula above, we have:

$$p - \rho ax + \rho gz = c$$

Considering $p = p_0$ on the surface, the slope of surface angle is :

$$k = \tan(\varphi) = \frac{z}{x} = \frac{a}{g}$$

Therefore, the slope of surface angle [see Figure 5] is proportional to acceleration a . We can devise an optimal scheme for the acceleration during the walking process to control the liquid sloshing, which is characterized by $d\varphi/dt$.

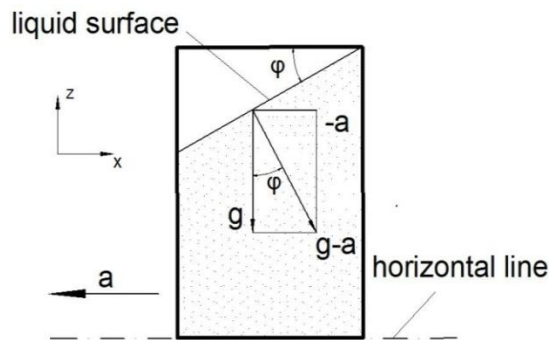


Figure 4. The stability of forces on the cup.

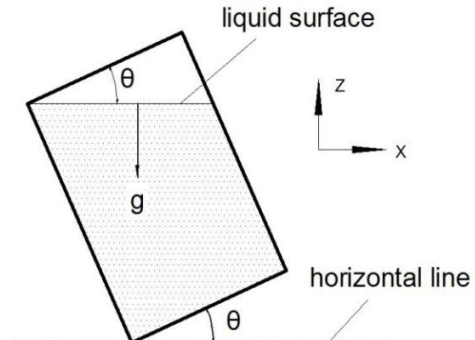


Figure 5. Postures of holding the cup.

4. Postures of holding the cup

When with a small cup tilt angle θ and the changing rate $d\theta/dt$, coffee in the cup can be regarded as stable. Thus we use θ and $d\theta/dt$ to characterize the postures of holding the cup during walking. We expect the surface sloshing as weak as possible, or say in a more mathematical way, we should keep the surface angle φ and the changing rate of it as small as possible all along. Theoretically, the cup tilt angle θ , which can be adjusted by the walker, need to change synchronically with the surface angle θ to hold more coffee in the container.

Schemes for the motion of a robot under different road condition

The real road condition can be treated as a combination of three basic types [see Figure 6]:

- condition1: a straight road;
- condition2: a road with upward and downward slopes;
- condition3: a road with corners.

Thus we can just model for these basic conditions to find a best motion scheme for the walking process.

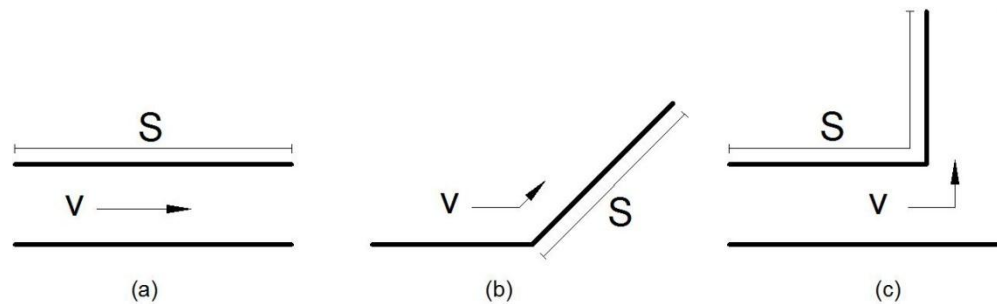


Figure 6. Basic road conditions.

Currently, the robot has mainly two kinds of walking mode: one is the wheeled mobile robot and the other is the biped robot walking with two mechanical feet. The biggest difference of the two types is the motion style, for the former walks smoothly and steadily compared with the latter's movement, which is more tottering. Consequently, it will generate great difference to the schemes for reducing the spill amount.

Our research starts from the easier type of robot, i.e. the wheeled mobile robot. The biped robot will be studied in model 3.

Model Solving

Model 1:an acceleration design based on quasi-static balance equation

Quasi static process is a process in which change in any of the parameters take

place at such a slow speed that the values of P, V, T can be taken to be practically constant. No real process is quasistatic, but such processes can be approximated by performing them very slowly^[3]. During the motion of coffee cup, the whole process can be considered as quasistatic providing that the acceleration a changes sufficiently slowly, in which case the liquid surface in the cup recover to the equilibrium state in a very short time.

As mentioned in problem 1, there are many factors accounting for the spill amount of coffee. However, if we regard the motion of the cup as a quasi-static process relevant with acceleration a and the cup tilt angle θ , all the influence above can be simplified to the influence of a and θ . From D'Alembert's principle, there is an imaginary inertia force [see figure 4] acting on the coffee in the cup[see Figure 7].

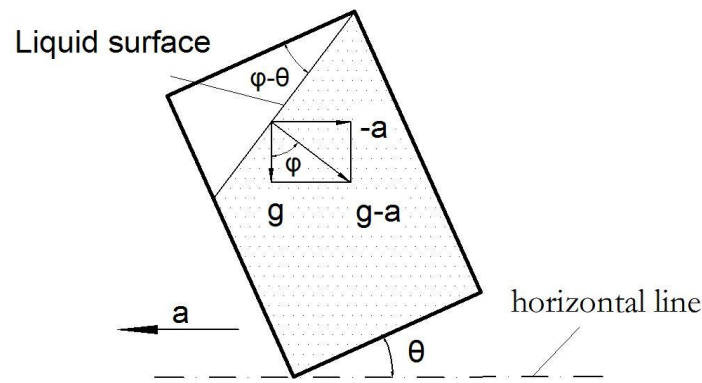


Figure 7. Quasi static process.

The coffee keeps a quasi static balance with gravity and inertia force. And the surface of coffee is always vertical to the resultant force of the cup, according to the geometric relationship shown in Figure 7, we find that the liquid sloshing with respect to the cup can be described by $d(\varphi - \theta)/dt$. Therefore, the spill amount relates to both da/dt and $d\theta/dt$ during the movement.

At the initial time, the cup is under static state, $\theta = 0$. To reduce the relative sloshing $d(\varphi - \theta)/dt$, we should minimize the initial acceleration $a|_{t=0}$ and let the changing rate be steady as possible, thus let $a|_{t=0} = 0$.

As the robot, starting from static state, needs to “walks rapidly”, acceleration is compulsory. There are many choices for designing the acceleration curve which can be concluded to four types of pattern [see Figure 8] in the following:

- The linear acceleration pattern like $a = K_1 t$;
- The radical acceleration pattern (accelerate rapidly at first then slowly) like $a = K_2 \sqrt{t} \ (t \leq 0.25T)$;

- The quadric acceleration pattern (accelerate slowly at first then rapidly) like $a = K_3 t^2$ ($t \leq 0.25T$);
- The circular acceleration pattern (accelerate slowly in the initial and final stage, while rapidly in the mid-term) like $a = K_4 - \sqrt{K_4^2 - t^2}$ ($t \leq 0.25T$).

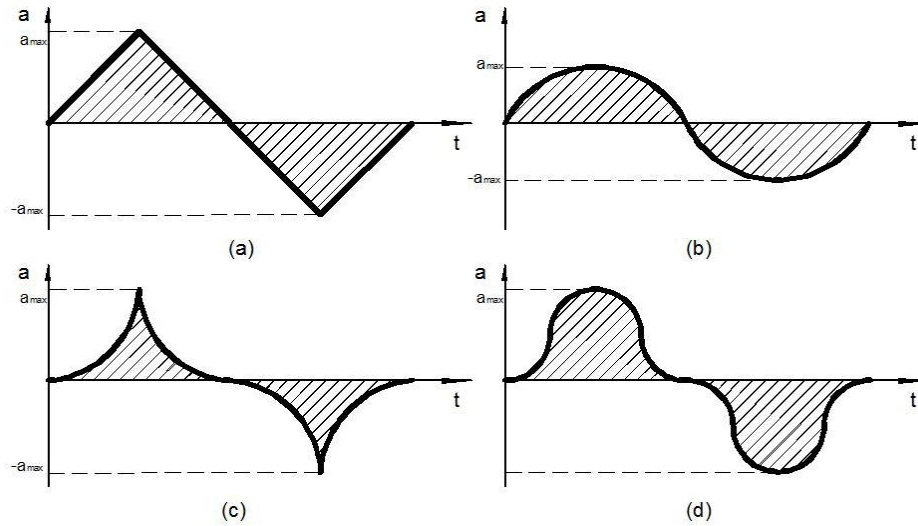


Figure 8. Different acceleration curves.

Our task is to find the best acceleration pattern among the four by simulating the walking process.

As is shown in Figure 9, the changing rate da/dt is smaller when the cup speed changes all along, compared with the speed changing locally. Therefore, for the four types of acceleration above, we let the cup speed change during the whole process for a smaller da/dt .

Taking the linear acceleration pattern as example, we can easily get the velocity curve by integrating acceleration. That is to say, the motion of a robot can be determined when we decide on one of the four acceleration types.

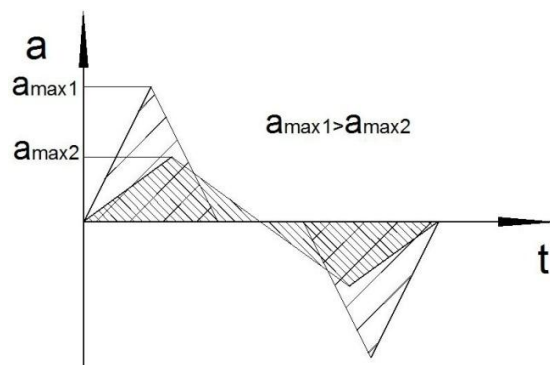


Figure 9. The reason for choosing the-whole-process change

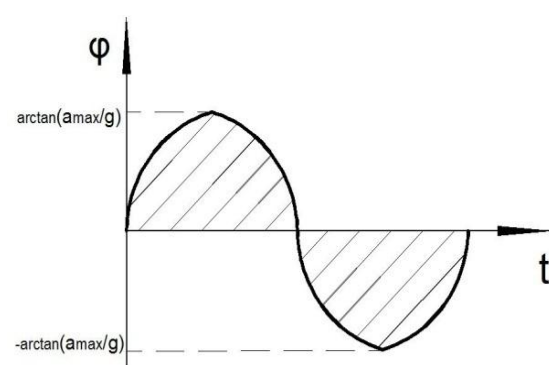


Figure 10. The velocity curve of the linear acceleration pattern.

The surface angle φ at any time is defined after we choose an acceleration type under the quasi static condition. We can minimize the spill amount by changing the cup tilt angle $\theta(t)$. It is easy to find that there is theoretically no spill amount when $\theta(t) = \varphi(t)$, namely the optimal movement effect. We have $\tan(\varphi(t)) = a(t)/g$ from Figure 9, so $\varphi(t) = \tan^{-1}(a(t)/g)$. Finally we can get the cup tilt angle curve $\theta(t)$. The $\theta(t)$ curve at the linear acceleration pattern is shown in Figure 10.

Model 2:the spherical pendulum model

Model 2 is based on the fully nonlinear spherical pendulum equations with forced oscillations. As a spherical pendulum (along with spring-mass models [see Figure 11]) has characteristics similar to sloshing.

The equivalent length of the pendulum is determined by the rotation frequency.

$$T = 2\pi \sqrt{\frac{l}{g}} = \frac{2\pi}{\omega}$$

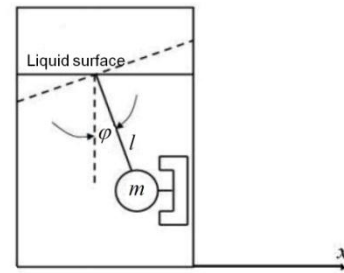


Figure 11. The spherical pendulum model and the equivalent length.

As shown in Figure 11, l is the equivalent length of the pendulum corresponding to the height of the coffee surface, m is the mass of coffee, φ is the vibration angle of the surface, x is the displacement of the cup and η is the viscosity of coffee. After some derivation, we have the system kinetic energy T :

$$T = \frac{1}{2} m \left(\dot{x} - l \dot{\theta} \cos \theta \right)^2$$

The system potential energy W is:

$$W = -mg(l - l \cos \theta)$$

The system energy loss F is:

$$F = \frac{1}{2} \eta \left(l \dot{\theta} \cos \theta \right)^2$$

According to lagrange equation,

$$\frac{\partial}{\partial t} \frac{\partial T}{\partial \dot{\theta}} - \frac{\partial W}{\partial \theta} + \frac{\partial F}{\partial \dot{\theta}} = 0 \quad (4)$$

using some simplified approaches referring to the literature^[5], we get the surface oscillation equation as follow:

$$\ddot{\theta} = \frac{\ddot{x}}{l} - \frac{\eta}{m} \dot{\theta} - \frac{g}{l} \theta \quad (5)$$

As mentioned in problem description, the robot walks “rapidly”. For model solving, we should define the “rapid walk” quantitatively: “walking rapidly” means that the robot must accomplish a distance S in a specified time t , which requires a minimal average velocity of a robot.

The radius of common cups ranges from 2.5 to 6.7 cm and the height of cups varies from 5.7 to 8.9 cm^[6]. For convenience, we choose $R = 3.5\text{cm}$ and $H = 10\text{cm}$, the equivalent length of the pendulum $l = 0.5 \times H$. The viscosity of coffee $\eta = 10 \sim 200 \times 10^{-3} \text{Pa} \cdot \text{s}$ ^[7], here let $\eta = 100 \times 10^{-3} \text{Pa} \cdot \text{s}$. The density of coffee $\rho = 1710 \text{kg/m}^3$. The acceleration of gravity $g = 9.8 \text{N/kg}$. We require that the robot should cover a distance of $S = 30\text{m}$ within $t = 10\text{s}$, which is a reasonable speed for the definition of “walking rapidly”.

Table 1.

parameter	value
R	3.5cm
H	10cm
l	$0.5 \times H$
η	$100 \times 10^{-3} \text{Pa} \cdot \text{s}$
ρ	1710kg/m^3
S	30m
t	10s
g	9.8N/kg

Firstly we study on the linear acceleration under the condition of a straight, road (condition 1, i.e.).

According to the surface oscillation equation (4) and the linear acceleration pattern [see Figure 6(a)], we have the variation of the surface angle φ with time [see Figure12].

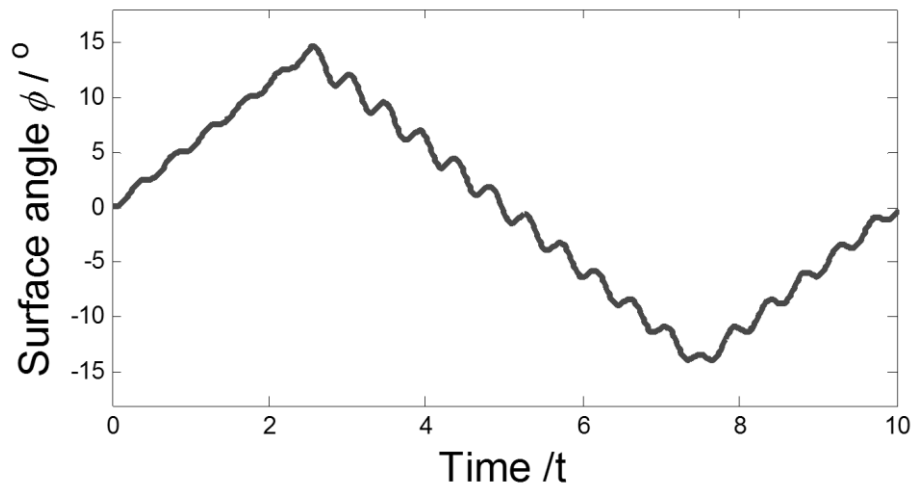


Figure 12. Variation of the surface angle ϕ with time in **condition 1**.

From the analysis of quasi static balance, the spill amount is optimal if $\theta(t) = \phi(t)$ is valid at any time. However, by the limitation of mechanical error, the motion of a robot cannot be controlled that accurately and consecutively. Therefore we have to pick out a simple curve that approximates most to $\phi(t)$. It is that simple curve that can be a design for cup tilt angle $\theta(t)$, which can be actually performed by a robot. The result of fitting, which is a piecewise linear function, is shown in Figure 12.

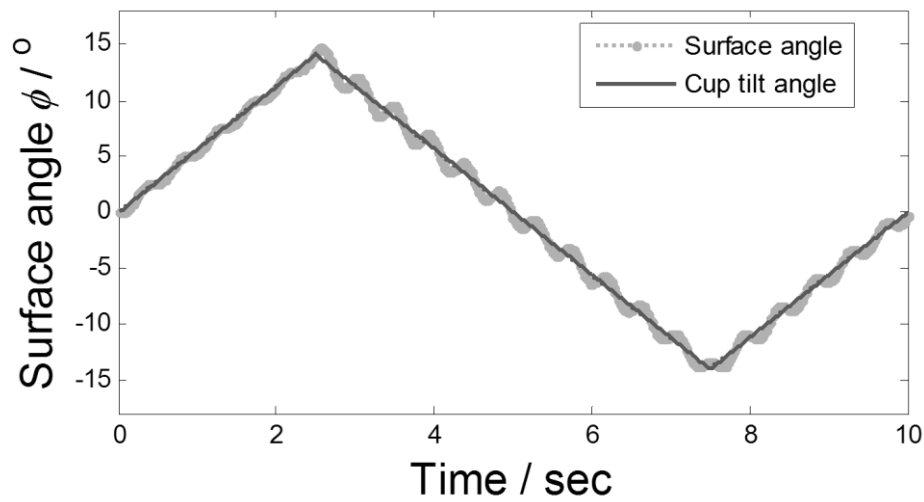


Figure 13. Fitting curve of the surface angle .

The analytic expression of the fitting curve.

$$\phi = \begin{cases} A_1 t + A_0, & 0 \leq t \leq 2.5s \\ B_1 t + B_0, & 2.5s < t \leq 7.5s \\ C_1 t + C_0, & 7.5s < t \leq 10s \end{cases}$$

Table 2. Results of the regression.

Line1 regress coefficient	Estimate value	Confidence interval
A1	9.839e-2	[9.748e-2 9.929e-2]
A0	-1.028e-3	[-2.344e-3 2.878e-4]
R ² = 0.9966 F=167 p<0.0001 s ² =0.0043		
Line2 regress coefficient	Estimate value	Confidence interval
B1	-9.822e-2	[-9.905e-2 -9.739e-2]
B0	0.4916	[0.4873 0.496]
R ² = 0.9941 F=327 p<0.0001 s ² =0.0112		
Line3 regress coefficient	Estimate value	Confidence interval
C1	9.708e-2	[9.531e-2 9.884e-2]
C0	-0.9721	[-0.9875 -0.9566]
R ² = 0.9862 F=165 p<0.0001 s ² =0.0085		

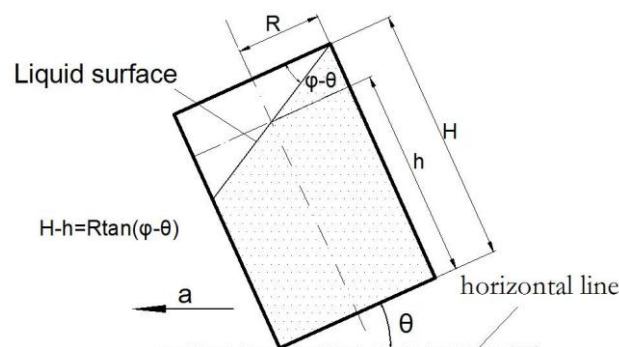
The fitting curve above is a practicable strategy for the cup tilt angle $\theta(t)$. Combining $\theta(t)$ and $a(t)$, we can get an ideal design for the motion of a robot.

Using computer technology, we performed a simulation of the motion of the cup. As all the disturbance factors can be concluded to the influence on acceleration a , we can use some influence coefficients to represent the disturbance.

Thus we have

$$a = a_0 \cdot K_r \cdot K_m \cdot K_\varepsilon$$

Where a is the actual acceleration performed in the motion, a_0 is the acceleration given by our designed acceleration curve, K_r is the impact factor of the road, K_m is the impact factor of mechanical transmission error, K_ε is the impact factor of random disturbance. According to the statistics knowledge, the total influence conducted by those impact factors obeys normal distribution. By consulting some reports^[4], we find that the deviation of a relative with a_0 is within 1% at a probability of 99.7%.

**Figure 14.** Liquid level ratio h/H and other parameters.

Then we simulated the motion with actual acceleration $a(t)$ and the design of cup tilt angle $\theta(t)$ for 1000 times. We calculated the height of the remaining coffee $h = \min\{H - R \cdot \tan(\varphi(t) - \theta(t))\}$ during each simulation. Let h be divided by the height of the cup, namely liquid level ratio h/H . We get the following results of 1000 times simulation.

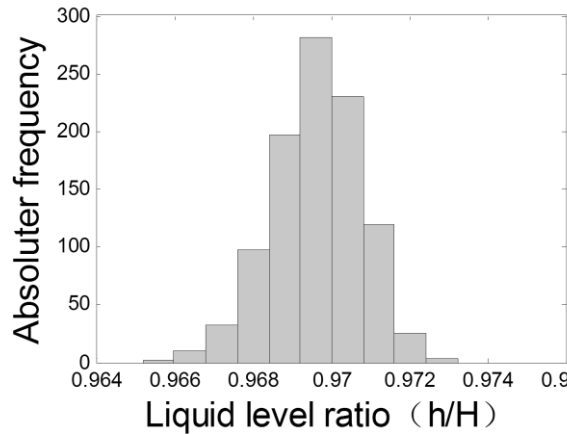


Figure 15. Result of 1000 tests.

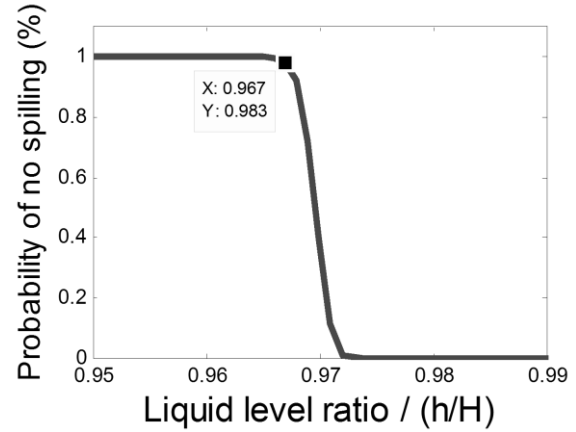


Figure 16. Probability of no spilling of different liquid level ratios.

Considering that there exist random factors, Figure 15 gives the frequency distribution of remaining liquid level ratio h/H . Figure 16 shows the probability of no spilling at different initial surface height h . Taking a point P on the curve as example, it means that the probability of no spilling is 98.3% when the initial surface height h is $0.967 \cdot H$.

Model 3: a good motion design for biped robots

The previous model shows that to the wheeled mobile robot, more than 97% of coffee can remain in the cup after the motion, which is too ideal a result compared with actual situation. To build a more accurate model, we research on the biped robots, whose walk is more complex.

By referring to the literature that analyze the force distribution of human's walking, we find that the force, which acts on the people when walking, is discontinuous. The real acceleration and deceleration of a biped robot is shown in Figure 16 and Figure (17) respectively.

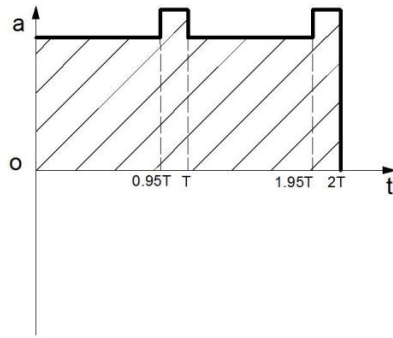


Figure 17. A biped robot's acceleration.

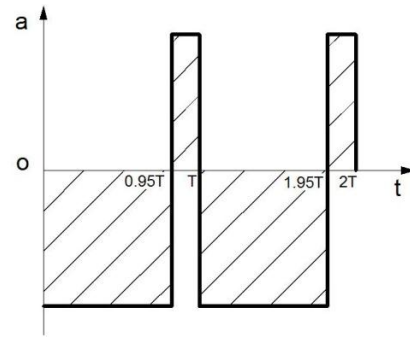


Figure 18. A biped robot's deceleration.

Modifying our previous model with the new acceleration curve, we get a new curve of $\varphi(t)$ in the following Figure 18.

Similarly we conduct the simulation of the motion for 1000 times again, the new probability distribution with different h/H is seen in Figure 20.

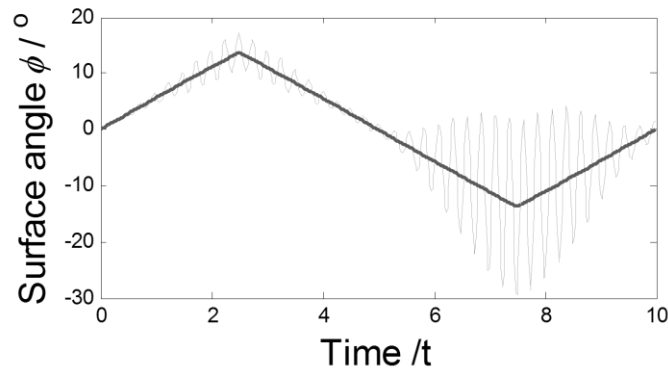


Figure 19. The real variation of the surface angle with time in **condition 1**.

This time we find that to the point P' on the curve, the probability of no spilling is 96.8% when the initial surface height h reduces greatly to $0.8909 \cdot H$. That is to say, when taking into account the disconnection of walking process, we get more spill amount of coffee, which is more reasonable to the actual conditions.

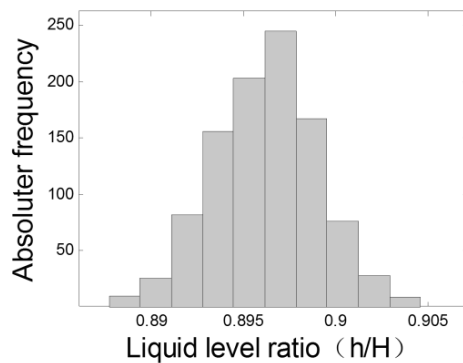


Figure 20. Result of 1000 tests.

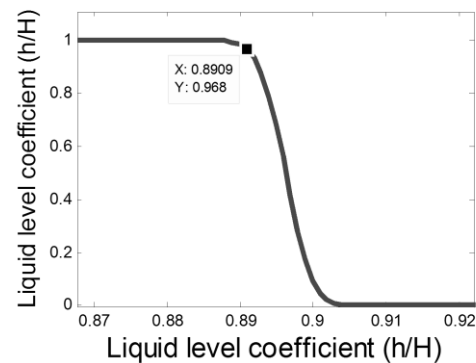


Figure 21. Probability of no spilling of different liquid level ratios.

As the model 3 built for biped robots is more reasonable than the previous models, we use model 3 to analyze other road conditions subsequently.

By the three models mentioned above, we have a quantitative analysis on the effect of liquid sloshing. Using model 3, we can get an ideal design of the cup tilt angle $\theta(t)$ when the acceleration of the robot $a(t)$ is given. By simulating the liquid sloshing for 1000 times, we can calculate the volume of coffee remaining in the cup. Thus we can minimize the coffee spill amount by choosing an optimal motion design $a(t)$.

As illustrated in model 1, mainly we have four types of acceleration plan: $a_1 = K_1 t$ 、 $a_2 = K_2 \sqrt{t}$ 、 $a_3 = K_3 t^2$ 、 $a_4 = K_4 - \sqrt{K_4^2 - t^2}$. In the following we will simulate the liquid motion under different road conditions and acceleration plans.

Model Applications

Comparison among four acceleration types on a straight, road (condition 1)

All the parameters are still the same with the previous [see in table 1]. According to the requirement of the minimal average velocity S/t , we can find the best acceleration curve under this road condition.

Table 3. Some parameters again.

Parameter	Value
R	3.5cm
H	10cm
l	$0.5 \times H$
η	$100 \times 10^{-3} Pa \cdot s$
ρ	1710 kg/m^3
S	30m
t	10s
g	9.8 N/kg

After simulating the four types of acceleration, we get their respective h/H , which represents the remaining amount of coffee. The results are as follow:

Linear acceleration type: $h/H = 0.8909$

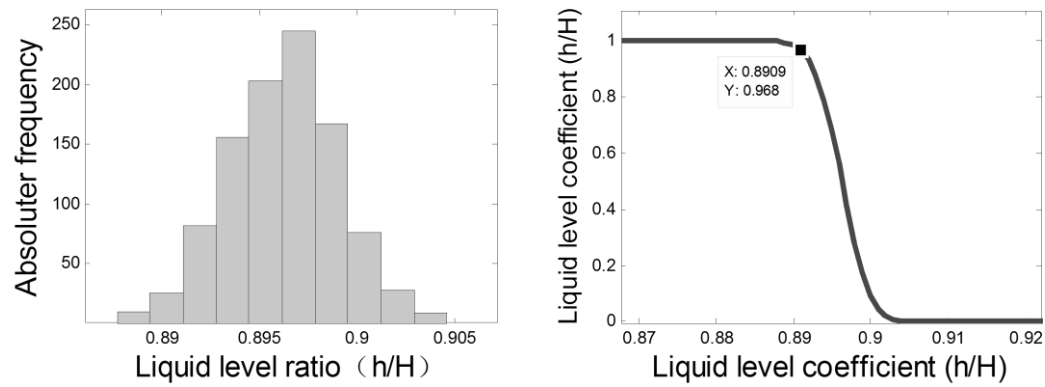


Figure 22. 1000 tests of linear acceleration type.

Radical acceleration type: $h/H = 0.8999$

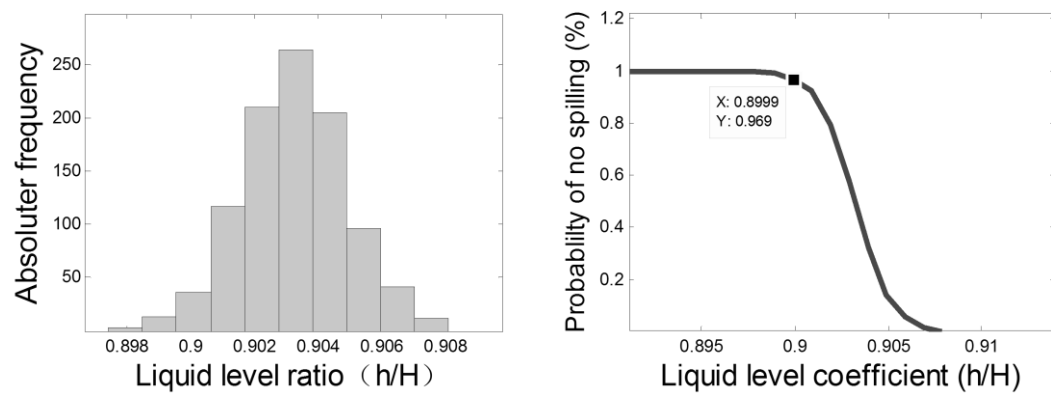


Figure 23. 1000 tests of radical acceleration type.

Quadric acceleration type: $h/H = 0.7988$

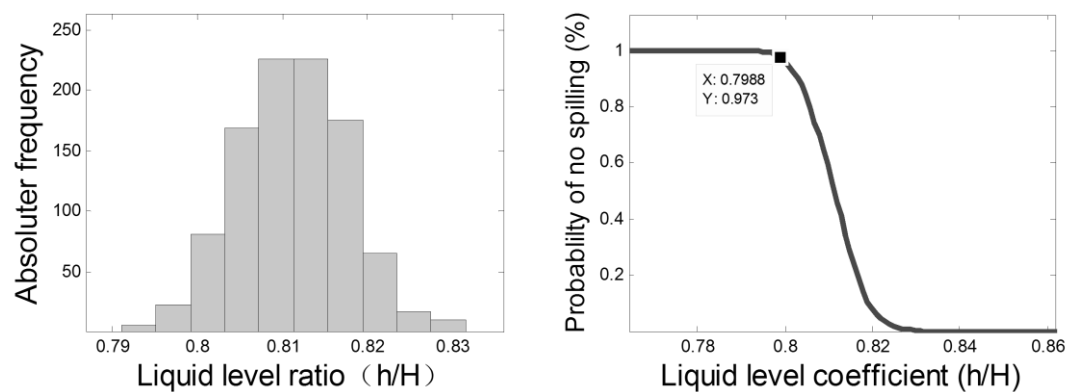


Figure 24. 1000 tests of quadric acceleration type.

Circular acceleration type: $h/H = 0.8008$

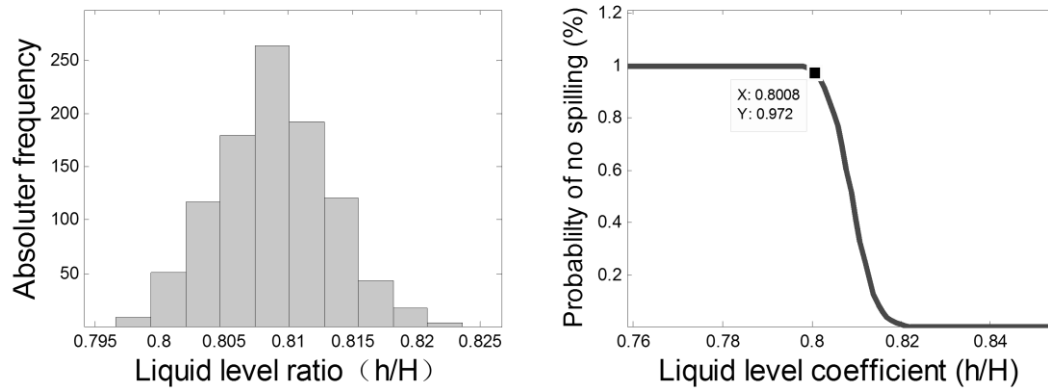


Figure 25. 1000 tests of circular acceleration type.

Accordingly, under the condition of a straight, road condition, the superiority of four acceleration types sorted from top to down is: radical type>linear type>circular type> quadric type. The difference between the radical type and the linear type is very slight.

Comparison among four acceleration types on a road with upward and downward slopes (condition 2)

The acceleration of the coffee cup on an upward slope is shown in Figure 26.

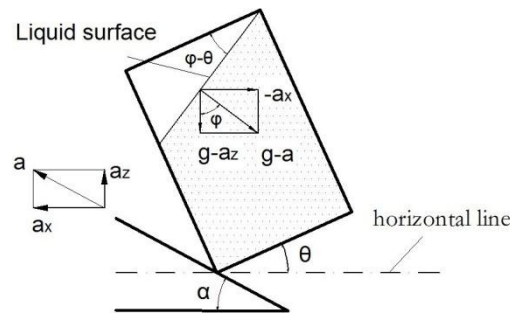


Figure 26. The acceleration on an upward slope.

According to the conclusion drawn in problem I, we hope the liquid surface sways slightly. Namely we want $\phi = \tan^{-1}[-a_x/(g - a_z)]$ (see 1) to approach to 0 as much as possible. Therefore, when going upwards, component of the acceleration on x-axis a_x and z-axis a_z should be as small as possible.

For convenience, we simplified the joint between the plane and the slope to a segment of arc. Then we divide the whole slope into five parts [see Figure 27]. To get a relatively small ϕ , we let a_x in T3 be 0. Our task is to choose acceleration for other time periods from the four acceleration types.

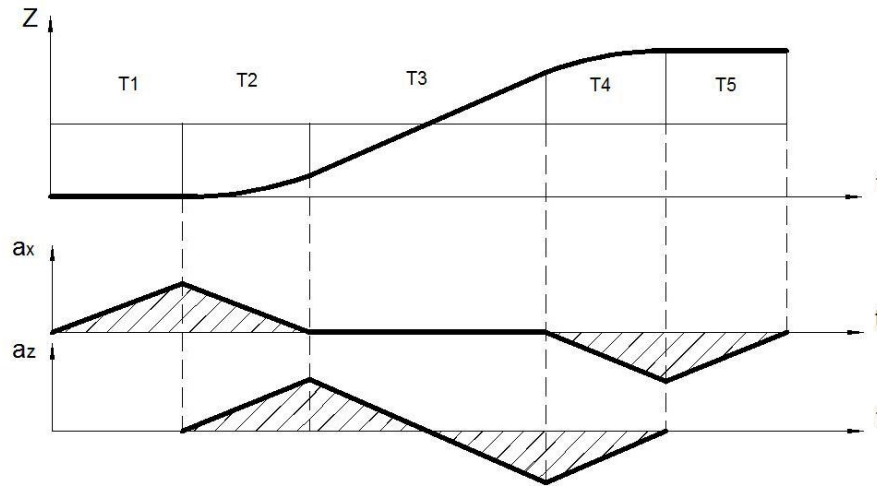


Figure 27. Five parts of the slope and their acceleration.

For simplification, let $T1=T2=T4=T5=T_c$, $T3 = \beta \cdot T_c$ ($\beta \geq 1$). From symmetry properties we know that the acceleration of T1, T2, T3, T4 are all equivalent. After some integrating, we have the following expressions:

$$k_{ax} = \frac{S \cos \alpha \cdot (4 + \beta)^3}{T^3 \left(\beta + \frac{5}{3} \right)}, 0 \leq t \leq T_c$$

$$k_{az} = \frac{16S \sin \alpha \cdot (4 + \beta)^3}{T^3 (\beta + 4)^3}, 0 \leq t \leq T_c$$

Where K_{ax} and K_{az} are the slope of the two component acceleration, S is the distance of motion, T is the total time and the α is the dip angle of the slope.

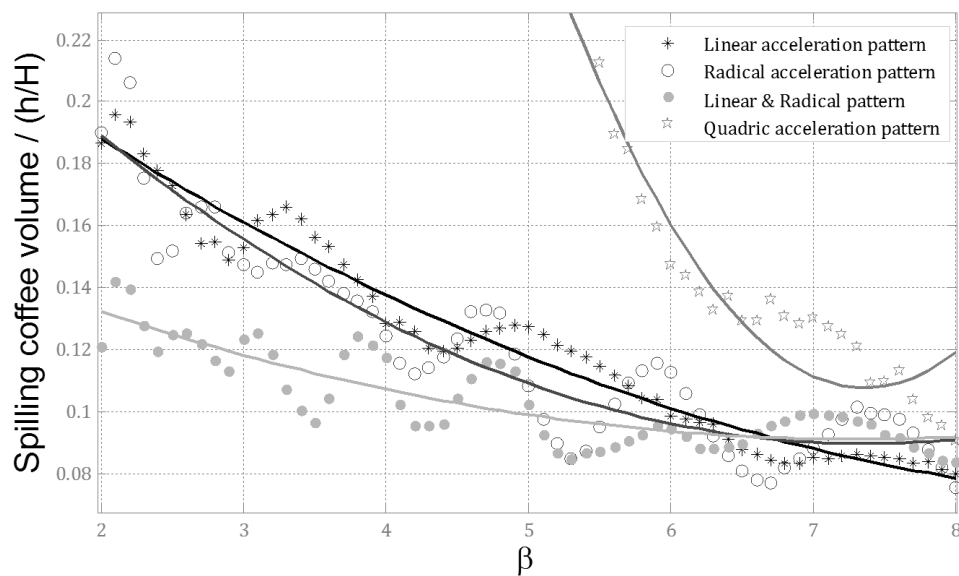


Figure 28. Liquid level ratios with different acceleration types and β .

Let $S = 30\text{m}$, $T = 10\text{s}$, we simulate the four acceleration types under the slope condition. Then change β from 2 to 5, and we will have the variation of the surface angle (t) , the cup tilt angle $\theta(t)$ as well as the liquid level ratio h/H .

Among the four acceleration types, when β is relatively small ($\beta \leq 6.7$), the radical acceleration type is optimal; When $\beta \geq 6.7$, the linear acceleration type is optimal.

Moreover, we surprisingly find that when we design a combination of the two optimal types, that is to say the acceleration on X-axis is radical while the acceleration on Y-axis is linear, we get a better simulation result than both the two types.

Comparison among four acceleration types on a road with corners (condition 3)

When making a turn with coffee, a robot has the acceleration on both the x direction and y direction in the rectangular coordinates system [see in figure 29]. φ is the surface angle in the XOZ plane. ψ is the angle describing the deviation of the pendulum in YOZ plan.

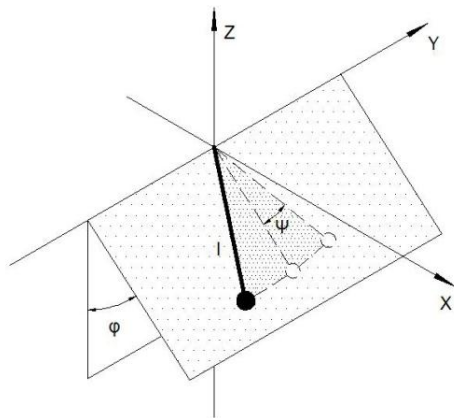


Figure 29. Definition of ψ and the location of the pendulum.

The transformation equation between the rectangular coordinates and the cylindrical coordinates are:

$$\begin{cases} x = l \cdot \cos \psi \sin \varphi \\ y = l \cdot \sin \psi \\ z = l \cdot \cos \psi \cos \varphi \end{cases}$$

Still, the system kinetic energy T , potential energy V and the energy loss T are as follows:

$$T = \frac{1}{2} m \left[\left(\dot{x} - l \dot{\theta} \cos \theta \right)^2 + \left(\dot{y} - l \dot{\varphi} \cos \varphi \right)^2 \right] \quad V = -mgl (1 - \cos \theta \cdot \cos \varphi)$$

$$F = \frac{1}{2} yl \left(\dot{\theta} \cos \theta + \dot{\varphi} \cos \varphi \right)^2 \quad \frac{\partial}{\partial t} \left(\frac{\partial T}{\partial \dot{\theta}} \right) - \frac{\partial W}{\partial \theta} + \frac{\partial F}{\partial \dot{\theta}} = 0$$

According to lagrange equation again, similarly we get:

$$\ddot{\theta} = \frac{\ddot{x}}{l} - \frac{\eta}{m} \left(\dot{\theta} + \dot{\varphi} \right) - \frac{g}{l} \theta$$

$$\ddot{\varphi} = \frac{\ddot{y}}{l} - \frac{\eta}{m} \left(\dot{\theta} + \dot{\varphi} \right) - \frac{g}{l} \varphi$$

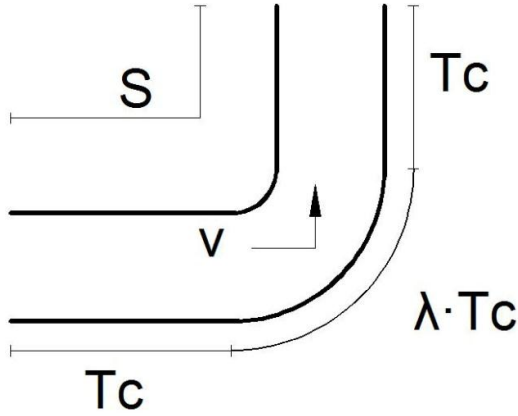


Figure 30. The situation when making a turn.

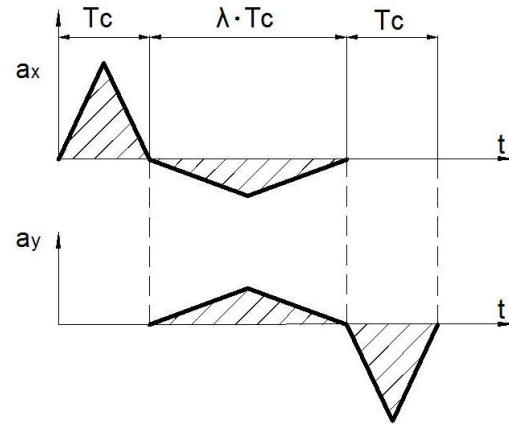


Figure 31. The variation of the acceleration.

Let $\gamma = 90$, and the robot enters in the x direction and leave in the y direction. Likewise, we hope that a_x and a_y both change slowly, a feasible linear acceleration type is seen in Figure 31. We divide the turning process into three parts: the entering stage, the turning stage and the leaving stage. λ is the time coefficient of proportionality. For convenience, we suppose the time of the entering stage is the same with that of the leaving stage. Using Runge-Kutta method, we get the result of the linear acceleration type.

Using similar methods, we study on the other three acceleration types. Letting $\lambda = 2 \sim 5$, we find that the linear and the radical acceleration pattern are insensitive with λ [see Figure 32], the spill amount of the two acceleration almost varies in a same range, while the linear acceleration type performs better (the spill amount is approximately $0.18 (h/H)$), the safe liquid level ratio is 0.82.

As for the quadric acceleration type, it shows a high sensitivity to λ and performs well when $\lambda > 4.3$ (the spill amount is about $0.10(h/H)$).

Therefore we suggest that we minimize the time spent on the entering and leaving stage, maximize the period in the turning stage (namely a relatively big λ), and choose the quadric acceleration type so that keep the spill amount as small as possible.

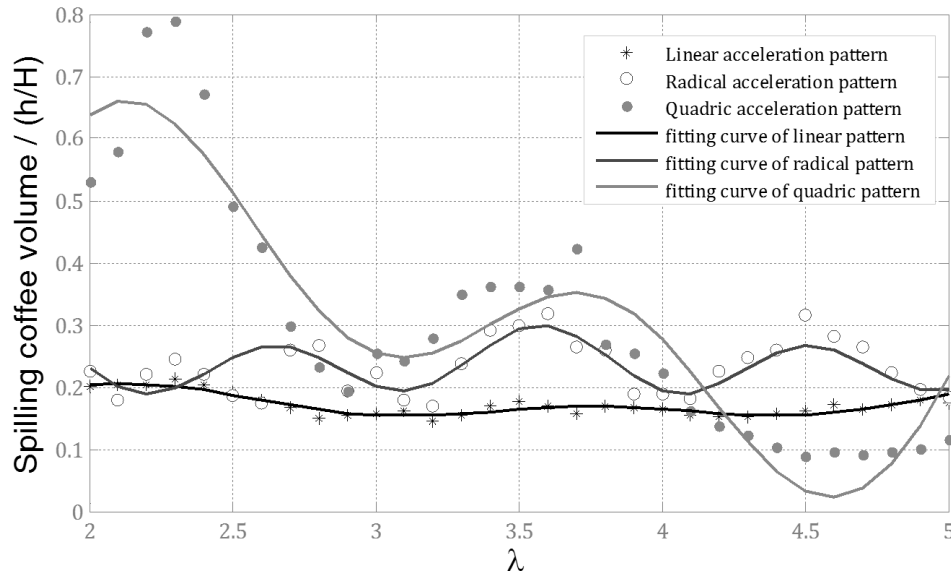


Figure 32. Liquid level ratios with different acceleration types and λ .

Optimal velocity choosing with different dimensions of cups

Through the analysis of problem 2, we are able to devise an ideal motion of robot (represented by $a(t)$ and (t)) and figure out the remaining amount of coffee in the cup under a certain road condition when the total distance and time is given.

When the spacing per step is determined, the frequency of driving force increase as the robot speeds up. The spill amount will be significantly different when involving the resonance effect. Therefore the velocity choosing becomes especially important in some circumstances where the resonance effect may be inspired.

Although the radical acceleration type is optimal, we choose the linear acceleration type for further research, as the difference of the two in terms of effect is not obvious.

Using the same parameter values mentioned in table 1 (or table 3), changing the average velocity v and setting $T = 10s$, we have the variation of the surface angle $\varphi(t)$ at different speed.

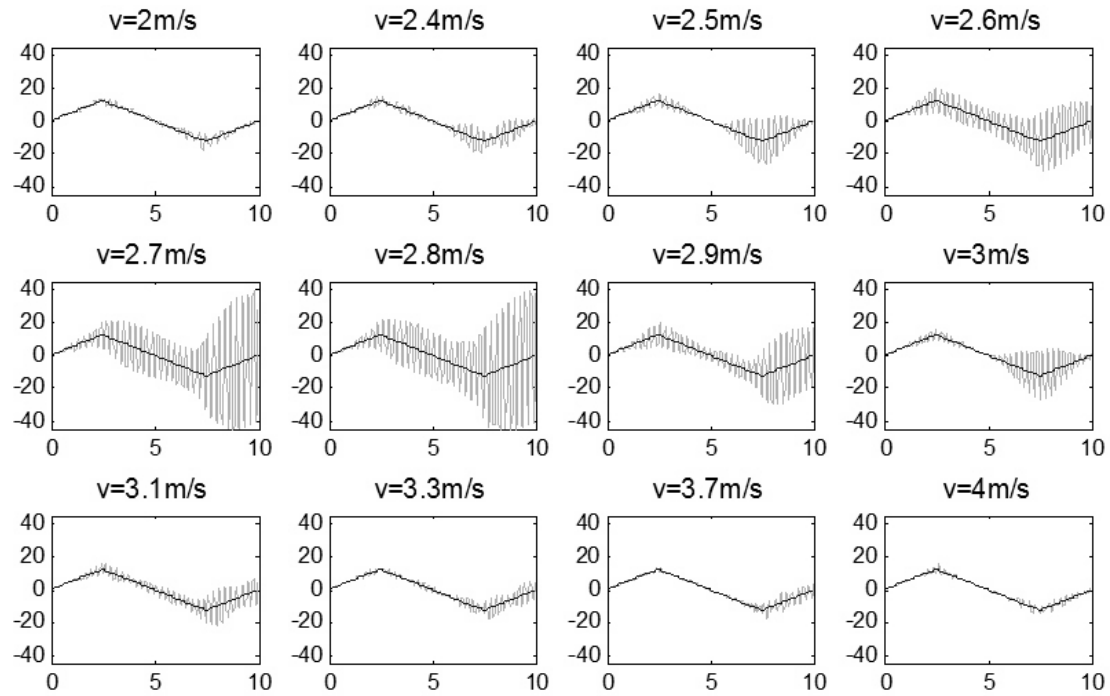


Figure 33. The variation of the surface angle at different speed.

The abscissa is the time of motion and the ordinate is the surface angle φ . The black polyline is the cup tilt angle $\theta(t)$ and the gray curve is the surface angle $\varphi(t)$. The more deviation between θ and φ , the less remaining amount of coffee is.

We get the most spill amount at the speed of 2.7m/s, and the frequency of walking step is 3.6Hz then, which is close to the nature frequency of cup (3.65Hz), calculated by equation 1.

From the analysis above, we get following discovery:

- The liquid sloshing increases significantly when the frequency of the driving force is close to the natural frequency of the cup. Therefore we must avoid the walking velocity which generates a frequency close to the natural frequency of the cup.
- The maximal liquid sloshing occurs in the decelerate stage, which is correspond to the actual situation.

Results

Concluding all the results and discovery during our modeling, we finally give the following conclusions:

- For the simulation of the straight, road condition, the radical acceleration pattern

keeps most coffee in the cup. According to the property of the radical curve, we suggest that when walking with coffee on the ground, one should accelerate swifter at the initial stage and adjust the cup tilt angle timely. Then the acceleration should change more and more slowly afterwards.

- For the simulation of the road with slopes, we suggest the speed of walker on the slope be as uniform as possible, so that he can have more coffee kept in the cup.
- For the simulation of the road with corners, we find that the quadric acceleration pattern keeps the most coffee. That means the acceleration of the walker should be small as possible when making a turn.
- From the comparison of different speed, we discovers that the velocity, which can generate a frequency far from the natural frequency of the cup, will perform the most remaining amount of coffee. So we should avoid the velocity that leads to the resonance effect.

Advantages

- We use the quasi static model and the spherical pendulum model to characterize the period and amplitude of liquid sloshing. These models are relatively simple, which are easy to realize the simulation.
- We analyze four acceleration types under three different road conditions, which expands the scope of model application, then we give suggestions to the actual situations.
- We analyze the influence of the walking frequency acting on the liquid sloshing, and give the reasonable velocity to the different dimensions of the cup.
- We regard the turbulence factor as a coefficient of velocity. That equivalent is helpful for the extension of our models.

Disadvantages

- The options for the acceleration are too few. We don't discuss a kind of acceleration pattern which has constant acceleration during some period of motion.

- We used some approximation when solving to simplify the equation. However, there does exist some situation where φ is relatively big. That contributes to the model distortion.
- There is some violent sloshing in the final deceleration period. We don't find a good way of controlling that great amplitude.
- We don't take into account the influence of the liquid surface tension as well as the radian of the liquid surface.

Model Extension

The mathematical models for describing the fluid flow consist of the continuity equation and the momentum balance equation.

To the incompressible fluid, we have the continuity equation

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

N-S equation:

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial P}{\partial x} + \rho g_x + \eta \left(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial P}{\partial y} + \rho g_y + \eta \left(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} + \rho g_z + \eta \left(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2} \right)$$

Since we research on the fluid flow in the cylindrical container, the equations above can be transformed into those at the cylindrical coordinates

$$\frac{\partial v_r}{\partial r} + \frac{v_r}{r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} = 0$$

and the corresponding N-S equation:

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta^2}{r} + v_z \frac{\partial v_r}{\partial z} \right) = -\frac{\partial P}{\partial r} + \rho g_r + \eta \left(\frac{\partial^2 v_r}{\partial r^2} + \frac{1}{r} \frac{\partial v_r}{\partial r} - \frac{v_r}{r^2} + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial^2 v_r}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} - \frac{v_\theta v_r}{r} + v_z \frac{\partial v_\theta}{\partial z} \right) = -\frac{1}{r} \frac{\partial P}{\partial \theta} + \rho g_\theta + \eta \left(\frac{\partial^2 v_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r^2} + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} + \frac{\partial^2 v_\theta}{\partial z^2} \right)$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial P}{\partial z} + \rho g_z + \eta \left(\frac{\partial^2 v_z}{\partial r^2} + \frac{1}{r} \frac{\partial v_z}{\partial r} + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right)$$

According to the four balance equations, we can realize the dynamic simulation of fluid flow and find the spill amount under different motion conditions.

References

- [1] H. Lamb, 1945. Hydrodynamics. Dover, New York.
- [2] H. F. Bauer, 1964, Technical Report No. TR R-187. NASA.
- [3] Quasistatic Process. Wikipedia. http://en.wikipedia.org/wiki/Quasistatic_process.
- [4] Li Kexuan. Error Analysis of Industrial Robots.
<http://www.doc88.com/p-213758814214.html>.
- [5] Zhang Yunhui, Shen Ying. 2010. The Application of Preshaping Method in Liquid Sloshing Control. Shen Yang. Journal of Shen Yang Ligong University. Vol 20. No 2.
- [6] H. C. Mayer and R. Krechetnikov. 2012. Walking with coffee: Why does it spill? California. Physical Review 85 E, 046117.
- [7] Approximate Viscosities of Some Common Liquids.
<http://www.michael-smith-engineers.co.uk/pdfs/ViscositiesofCommonLiquids2.pdf>.

Appendix

One of our matlab programs.

```
%机器人随机扰动模型一
clc,clear;
h=10/100;
r=3.5/100;
g=9.8;
w=23;
l=g/w/w;
eta=100e-3;%咖啡粘度系数
pho=1710;%咖啡密度
m=pi*r*r*h*pho;
tt=10;
s=30;
pace=0.75;
k=32*s/tt/tt/tt;
%k=192*s/tt/tt/tt/tt;%二次
%k=12*s/tt/tt/sqrt(tt);%根号
%k=solve(' (1-4*30/R/100)^2+(10/8/R)^2=1');k=subs(k);圆弧
%k=2*3.14*s/tt/tt;%正弦
loch=zeros(1000,1);
for j=1:1000
    %figure
    %for s=10:0.2:40
        N=500;
        x=zeros(N,1);
        y=zeros(N,2);
        p=tt/N;
        x(1)=0;y(1,1)=0;y(1,2)=0;
        for i=1:N
            ii=i;
            k1=p*pendulumrnd(x(ii),y(ii,1),y(ii,2),k,tt,l,g,m,eta,s,pace);
            k2=p*pendulumrnd(x(ii)+p/2,y(ii,1)+p*y(ii,2)/2,y(ii,2)+k1/2,k,tt,l,g,m,eta,s,pace);
            k3=p*pendulumrnd(x(ii)+p/2,y(ii,1)+p*y(ii,2)/2+p*k1/4,y(ii,2)+k2/2,k,tt,l,g,m,eta,s,pace);
            k4=p*pendulumrnd(x(ii)+p,y(ii)+p*y(ii,2)+p*k2/2,y(ii,2)+k3,k,tt,l,g,m,eta,s,pace);
            x(ii+1)=x(ii)+p;
            y(ii+1,1)=y(ii,1)+p*y(ii,2)+p/6*(k1+k2+k3);
            y(ii+1,2)=y(ii,2)+(k1+2*k2+2*k3+k4)/6;
        end
        plot(x,y(:,1)*180/pi);
        temp1=k*x.*(x<tt/4)+(-k*x+k*tt/2).*(x>=tt/4&x<3/4*tt)...
```

```

+(k*x-k*tt).*(x>=3/4*tt);

theta=atan(temp1/g);%直线
temp2=(k.*x.*x.*(x<tt/4)+k.*(x-tt/2).*(x-tt/2).*(x>=tt/4&x<tt/2)...
-k.*(x-tt/2).*(x-tt/2).*(x>=tt/2&x<3/4*tt)-k.*(x-tt).*(x-tt).*(x>=3/4*tt));
theta=atan(temp2/g);%二次
temp3=k.*sqrt(x).*(x<tt/4)+k.*sqrt(-x+tt/2).*(x>=tt/4&x<tt/2)...
-k.*sqrt(x-tt/2).*(x<3/4*tt&x>=tt/2)-k.*sqrt(-x+tt).*(x>=3/4*tt);
theta=atan(temp3/g);%根号
temp3=(k-sqrt(k^2-x.^2)).*(x<tt/8)+(sqrt(k^2-(x-tt/4).^2)+k-sqrt(4*k*k-tt*tt/16)).*(x>
=tt/8&x<tt*3/8)...
+(k-sqrt(k^2-(x-tt/2).^2)).*(x<tt/2&x>=tt*3/8)+(sqrt(k^2-(x-tt/2).^2)-k).*(x>=tt/2&x<tt
*5/8)+...
(sqrt(4*k*k-tt*tt/16)-k-sqrt(k^2-(x-tt*3/4).^2)).*(x>=tt*5/8&x<tt*7/8)+(sqrt(k^2-(x-tt
).^2)-k).*(x>=tt*7/8);

theta=atan(temp3/g);%圆弧
temp3=k*sin(2*pi/tt*x);
theta=atan(temp3/g);%正弦
theta=(0.09839*x-0.001028).*(x<=tt/4)+(-0.09822*x+0.4916).*(x>tt/4&x<=tt*3/4)...
+(0.09708*x-0.9721).*(x>=tt*3/4);

figure(2)

hold on

plot(x,theta*180/pi);

xlim([0 tt])

figure

plot(x,(theta-y(:,1))*180/pi);

loc=max(tan(abs(y(:,1))-theta*(1+normrnd(0,1/100)))));

V=(pi*r*r*r*loc);

disp(loc*r/h)

loch(j)=loc*r/h;

end

figure(3)

hist(1-loch);

M=1000;

pro=zeros(M,1);

safe=linspace(0,1,M);

for l=1:M

    pro(l)=(sum((1-loch>safe(l))))/1000;

end

figure(4)

plot(safe,pro,'m');

%xlim([0.830 0.890])

ylim([0,1.1])

```